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1 Project Charter - TravelHunters

1.1 Problem Definition

Domain: Travel and Tourism Data Aggregation

Problem Statement: Travelers today face information overload when planning trips, with relevant data scattered across multiple platforms (booking sites, travel guides, activity platforms). There is no centralized system that provides comprehensive, up-to-date travel information including accommodations, activities, and destinations with pricing, ratings, and visual content.

Expected Benefits: - Centralized travel data repository for better decision-making - Automated data collection reducing manual research time - Comprehensive dataset enabling travel analytics and recommendations - Foundation for future travel recommendation systems

Target Users: - Travel enthusiasts seeking comprehensive trip planning information - Travel agencies requiring aggregated data for client recommendations - Data analysts researching travel trends and pricing patterns - Developers building travel-related applications

Stakeholders:

- **Primary Users:** Travel planners and tourism professionals
- **Data Providers:** Booking.com, GetYourGuide, TripAdvisor, Lonely Planet, Google, Duckduckgo, activity platforms
- **Development Team:** Data engineers, web scrapers, database administrators
- **End Beneficiaries:** Travelers seeking better trip planning tools

1.2 Situation Assessment

Available Resources:

- **Personnel:** Development team with expertise in web scraping, data processing, and database management
- **Tools:** Python ecosystem (Scrapy, SQLite, Pandas, selenium, DDG_search), Quarto for documentation
- **Infrastructure:** Local development environment with capability for web scraping
- **Data Sources:** Public travel websites (Booking.com, TripAdvisor, Lonely Planet, Google Maps/Places, Get your Guide, Duckduckgo)

Time Constraints: - Project duration: July 2025 (Summer School timeframe) - Limited to proof-of-concept and initial data collection

Restrictions and Constraints:

- **Legal:** Must comply with website terms of service and robots.txt
- **Technical:** Rate limiting to avoid overwhelming target websites
- **Data Quality:** Dependent on source website structure and content quality
- **Scalability:** Limited by computing resources for large-scale scraping

Identified Risks: - Website structure changes affecting scraper reliability - Rate limiting or IP blocking by target websites - Data quality inconsistencies across different sources - Storage limitations for large datasets with images

1.3 Project Goals and Success Criteria

Primary Objectives: 1. **Data Collection Success:** Collect comprehensive travel data from multiple sources - Target: >2,000 hotel listings with pricing and ratings - Target: >500 activity listings with descriptions and images - Target: >200 destination entries with coordinates and descriptions

2. **Data Quality Standards:**

- 90% of entries have complete name and location information
- 80% of hotels have pricing information
- 70% of activities have rating information
- 60% of entries include image URLs

3. **System Reliability:**

- Scrapers handle pagination and multiple pages successfully
- Robust error handling for website changes
- Data cleaning pipeline removes duplicates effectively

4. **Database Integration:**

- All data successfully stored in structured SQLite database
- Proper indexing for efficient queries
- Data integrity constraints maintained

Success Metrics:

- **Quantity:** Total items collected across all categories
- **Coverage:** Number of unique destinations covered
- **Completeness:** Percentage of fields populated per category
- **Accuracy:** Manual validation of sample data entries

Out of Scope: - Real-time data updates or live synchronization - Advanced recommendation algorithms - User interface development - Commercial deployment or monetization

1.4 Data Mining Goals

Primary Data Mining Task: Data Integration and Information Extraction

Technical Objectives: 1. **Web Scraping and Data Extraction:** - Extract structured data from unstructured web content - Handle dynamic content and pagination - Parse and normalize pricing, rating, and location data

2. **Data Cleaning and Preprocessing:**

- Remove duplicate entries based on name/location similarity
- Standardize pricing formats and currency conversions
- Validate and clean URLs, ratings, and text content

3. **Data Integration:**

- Merge data from multiple sources (hotels, activities, destinations)
- Create unified schema across different data types
- Establish relationships between accommodations, activities, and destinations

4. **Feature Engineering:**

- Generate unique identifiers for entities
- Extract location coordinates where available
- Categorize and tag content appropriately

Quantitative Success Criteria:

- **Data Completeness Score:** >75% for critical fields (name, location, price/rating)
- **Duplicate Reduction:** <5% duplicate entries after cleaning
- **Data Integration Success:** Successfully merge >95% of collected data into unified schema
- **Processing Accuracy:** <2% data corruption during cleaning and transformation

Technical Requirements: - Scalable data processing pipeline handling 10,000+ records - Robust error handling with <1% data loss due to processing errors - Database performance supporting sub-second queries on full dataset

1.5 Project Plan

TravelHunters Data Collection and Integration Project

The project follows the CRISP-DM methodology adapted for web scraping and data integration:

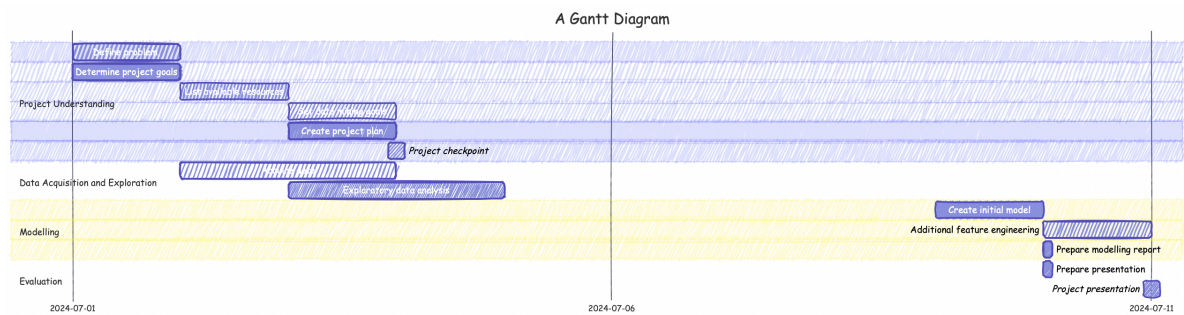


Figure 1.1: TravelHunters Project Timeline

```
gantt
    title TravelHunters Project Timeline - July 2025
    dateFormat YYYY-07-DD
    tickInterval 1day
    section Project Setup
        Define problem and goals      :done, a1, 2025-07-01, 1d
        Setup development environment  :done, a2, 2025-07-01, 1d
        Initial spider development     :done, a3, 2025-07-02, 1d
        Project foundation: milestone, done, m1, 2025-07-03, 0d
    section Data Acquisition
        Booking.com spider development :done, a4, 2025-07-02, 2d
        Activities spider implementation :done, a5, 2025-07-03, 1d
        Destinations data collection :done, a6, 2025-07-03, 1d
        Worldwide data scraping :done, a7, 2025-07-04, 2d
        Data collection complete: milestone, done, m2, 2025-07-06, 0d
    section Data Processing
        Data cleaning pipeline :active, a8, 2025-07-03, 2d
        Database integration :a9, 2025-07-03, 1d
        Duplicate removal and validation :a10, 2025-07-04, 1d
        Quality assurance :a11, 2025-07-04, 1d
    section Analysis & Documentation
        Exploratory data analysis :a12, 2025-07-05, 2d
        Documentation completion :a13, 2025-07-06, 2d
        Final evaluation :a14, 2025-07-07, 1d
        Project completion : milestone, m3, 2025-07-08, 0d
```

Phase Descriptions:

- **Project Setup (Days 1-3):** Environment setup, initial spider development, proof of concept
- **Data Acquisition (Days 2-6):** Large-scale data collection from multiple sources with pag-

ination

- **Data Processing (Days 3-10):** Cleaning, integration, and database storage
- **Language Model Construction & Training (Days 10-14):** Building Language Model, training & tweaking with textual data
- **Convolutional Neural Network Construction and integration (Days 15-19):** Building CNN, training with img datasets, integration with Language model
- **Graphical User Interface (Days 16-26):** GUI creation for smooth and streamlined usage